Gatsby Computational Neuroscience Unit, UCL

Advances in Adaptive Experimentation

Learning, decision-making, and inference under sequential data collection

Dates	June 18–19, 2026
Venue	Gatsby Computational Neuroscience Unit, UCL
Intro	Houssam Zenati, on behalf of the organizers

The object is the data-collection rule

- At round t: observe context and history, choose an action, observe an outcome.
- The design is history-dependent:

$$A_t \sim g_t(\cdot \mid H_{t-1}, X_t).$$

- The experiment is a stochastic process, not just a sample.

$$p(H_T) = \prod_{t=1}^{T} p_X(X_t) \times g_t(A_t \mid H_{t-1}, X_t) \times p_Y(Y_t \mid A_t, X_t).$$

The randomization rule is part of the statistical object.

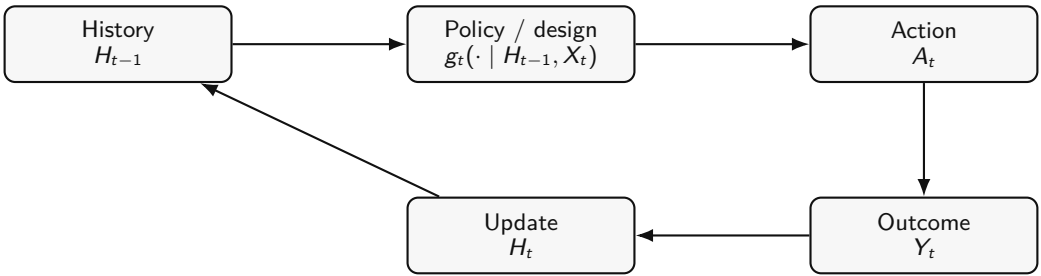
Why adapt? Three objectives compete

- **Identification:** learn the best arm, treatment, policy, or design with fewer observations.
- **Welfare/regret:** avoid repeatedly assigning units to actions that are already unlikely to be best.
- **Inference:** make statements whose uncertainty accounting respects the data-dependent design.

adaptive experimentation $\neq$ optimization only $\neq$ classical randomization only.

The tension is between learning fast, acting well, and still knowing what the estimates mean.

Feedback, not independent sampling



- Randomization probabilities are predictable, but not fixed in advance.
- Early noise changes later allocation.
- Later data therefore carry traces of earlier estimation error.

Where the CLT argument changes

- In a fixed randomized experiment, a Wald interval is usually backed by an i.i.d. or triangular-array CLT.
- Under adaptive allocation, asymptotically linear estimators typically look like martingale sums:

$$\widehat{\psi}_T - \psi = \frac{1}{T}\sum_{t=1}^{T} D_t + o_p(T^{-1/2}), \qquad \mathbb{E}[D_t \mid \mathcal{F}_{t-1}] = 0.$$

- The key quantity becomes the predictable quadratic variation

$$V_T = \sum_{t=1}^{T} \mathbb{E}[D_t^2 \mid \mathcal{F}_{t-1}].$$

- If V_T/T does not stabilize, the limiting law need not be a single normal law.

A minimal failure mode: explore–then–commit

- Explore uniformly for t_0 rounds.
- Commit mostly to the empirically better arm.
- If early noise favors arm 1, later arm-1 information is abundant.
- If early noise favors arm 2, later arm-1 information is scarce.

$$\sqrt{T}(\widehat{\psi}_T - \psi) \Rightarrow \begin{cases} N(0, V_1), & \text{early path 1,} \\ N(0, V_2), & \text{early path 2.} \end{cases}$$

$$\mathcal{L}_\infty = \pi N(0, V_1) + (1 - \pi) N(0, V_2).$$

The design can leave a first-order trace in the limit distribution.

What this workshop is about technically

- **Design:** exploration–exploitation, best-arm identification, structured feedback.
- **Stability:** regularization and conditions under which adaptive designs behave predictably.
- **Inference:** confidence sequences, betting, simulation-based inference, post-adaptive uncertainty.
- **Causality:** policy values, treatment effects, nuisance robustness, semiparametric efficiency.
- **Learning theory:** regret, transport, RL, sequential testing.
- **Open problems:** cases where learning objectives and inferential objectives still do not align.

The common question: how should data be collected when collection itself learns?

Programme: Thursday, June 18

Time	Session	Title / note
09:00	Registration and coffee	
09:30–09:45	Introduction	Organizers
09:45–10:45	Tor Lattimore	Continuous-time as a tool for understanding in RL, ML and beyond
10:45–11:00	Coffee break	
11:00–12:00	Aurélien Bibaut	Fast best-in-class regret for contextual bandits
12:00–13:30	Lunch	
13:30–14:30	Gergely Neu	Generative modeling by value-driven transport
14:30–14:45	Coffee break	
14:45–15:45	Emma Brunskill	Structured exploration for contextual bandits and RL
15:45–16:15	Coffee break	
16:15–17:15	Open problems	Short presentations and discussion

Programme: Friday, June 19

Time	Session	Title / note
09:30–09:45	Coffee	
09:45–10:45	Koulik Khamaru	Stabilizing bandit algorithms via regularization
10:45–11:00	Coffee break	
11:00–12:00	Kelly Zhang	Bandit simulation for average reward inference
12:00–13:30	Lunch	
13:30–14:30	Ian Waudby-Smith	Multi-armed sequential hypothesis testing by betting
14:30–14:45	Coffee break	
14:45–17:30	Poster session	Accepted posters

Practical information and acknowledgements

Practical

- WiFi: connect to `UCLGuest`. Instructions are posted at the entrance of the ground-floor seminar room.
- Photos: if you do not want to appear in photos, speak to one of the organizers.
- Open problems: concise pitches; leave time for technical discussion.
- Posters: Friday afternoon, 14:45–17:30.

Thanks

- Speakers and poster presenters.
- Participants.
- Gatsby, UCL, ELLIS Unit London, and CSML.
- Netflix for sponsorship.

The point of the next two days is to sharpen the interface between online learning, causal inference, and valid post-adaptive inference.